

Deep Inelastic Scattering Revisited

Felix Hekhorn
University of Jyväskylä

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Abstract

Discuss Yadism [\[1\]](#).

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Chapter 1

Introduction

1.1 Introduction

- Ich + paper + yad
- Ziele und $\neg$ Ziele
- D I S
- factorization theorem
- LO (+ NLOq + NLOg)
- IC

1.2 SF and xs

- x, Q^2
- $L_{\mu\nu} W^{\mu\nu}$

1.3 RSL

1.4 More particles

- γ/Z
- leptons
- hadrons: p+d+A; nucl. PDF
- experimental overview

1.5 DGLAP

- resummation
- Mellin

1.6 Scale variations

- scheme A/B/C
- MHOU
- covariance matrix

1.7 Heavy Quarks

- F_2^c
- FONLL

1.8 γ_5

- helicity

- HVBM + Larin

1.9 Positivity

- DIS vs. $\overline{\text{MS}}$

1.10 Higher orders

- IHOU
- N³LO
- channels
- loops and legs

1.11 QED

- Qiu et al.

1.12 Valo

- $C_\gamma \rightarrow k$

Bibliography

- [1] A. Candido, F. Hekhorn, G. Magni, T. R. Rabemananjara, and R. Stegeman, Eur. Phys. J. C **84**, 698 (2024), 2401.15187.